

A RECONSTRUCTION PROGRAMME IN STRUCTURAL MATHEMATICS

UNIFIED MECHANICS

Final Edition

Coherent Master · 2026.7 × Closing Addendum · 2026.8

φ

$$\varphi^2 = \varphi + 1, \quad r = \frac{1}{2\varphi}$$

JOSEPH SHIELDS

ONE INVARIANT · MANY SECTORS · ONE CONSISTENCY TEST



ABOUT THIS EDITION. This volume is the reading edition of the Unified Mechanics programme as it stands at the close of the 2026 sessions. Part I distils the governing core of the 2026.7 *Coherent Master Edition* — the statement of the programme, the operational chain, the golden dictionary, and the four-anchor representation boundary. Part II incorporates, in full, the 2026.8 *Closing Addendum* on cross-sector rigidity, identifiability, and falsification. The complete technical development (Chapters 1–11 and Appendices A–V of the master edition) remains the archival record; where the two differ in grading, Appendix V of the master and the ledger of Part II govern.

READING PROTOCOL. Preserve the grade attached to every claim — *Proved*, *Conditional*, *Conjectured*, *Proposed*, or *Postulate*. Distinguish the dated development record from the governing current ledger. Check proofs rather than slogans.

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one invariant is meaningful only when
many sectors cannot choose it independently



Part I

The Programme

Statement of the Programme

Unified Mechanics is a mathematical reconstruction programme whose objective is to determine whether the structures ordinarily taken as primitive — geometry, observables, variational principles, and continuum structure — are instead canonical consequences of a single operational foundation.

That sentence is the invariant of this document. Every chapter beneath it may be revised, strengthened, or refuted without changing it, because the sentence describes the question the programme investigates, not any answer it currently defends. A summary that does not preserve this abstraction is a misdescription.

1.1 What this document does not claim

To preclude the most common misreadings, the following are stated at the outset and are checkable against the text.

It does not claim an unconditional completed theory of everything. It contains a proved categorical and reconstruction core, explicit model classes, and a conditional physical closure theorem. The master closure result (Theorem 4.1 below) derives the downstream physical structure only after Postulate A, Anchors A1–A4, and one action calibration are stated openly.

It does not claim that rewriting alone derives cosmology. The cosmological formulas of §3.2 are promoted only conditionally by the relational registration machine and the common stress-energy calibration; the physical realization of that readout remains testable input, not a theorem of the rewriting axioms.

The golden ratio is not the thesis. φ enters once, as the seed of a contraction algebra; the load-bearing content is the reconstruction machinery — quotients, universal properties, and factorization theorems — which is independent of any numerological reading.

Elegance is not offered as evidence. The programme's own critical assessment states the central criticism — inevitability claimed where uniqueness was unproven — more precisely than any external review has, and concedes it in full.

1.2 Minimum description

If one sentence of this document is to survive in a reader's memory, it should be this: *Unified Mechanics is a structural reconstruction programme with a proved categorical core and an explicit conditional physical realization, whose remaining content is concentrated into four auditable representation anchors and one dimensional calibration.*

The System Before the Physics

The logic of this programme should be followable before any reader meets the cosmology. The starting idea is not that a word becomes a particle or that an equation becomes a galaxy. The starting idea is that a system capable of describing itself must distinguish what changes with the description from what remains invariant.

2.1 Descriptions and what survives them

Let C be a set of admissible presentations and let $\rightarrow$ be an allowed reduction. A presentation is not yet the thing itself; it is one way the system can be written, executed, or inspected. Self-description motivates an invariance requirement: admissible changes of presentation should not change the completed content. To turn that motivation into a theorem, the rewrite system must satisfy explicit hypotheses. *Termination* says that reduction does not continue forever. *Local confluence* says that two immediate reductions from the same presentation can later be joined. Under those two hypotheses, Newman's lemma gives a unique normal form $\kappa(X)$ for every presentation X . This is the first rigorous compression: many admissible routes lead to one invariant result.

Define $X \sim Y$ exactly when $\kappa(X) = \kappa(Y)$. The quotient $F = C/\sim$ removes distinctions the system cannot preserve under complete reduction. Any observable constant along reduction factors through this quotient; if a re-presentation group acts equivariantly, κ respects that action.

2.2 Why three terms appear

The quotient theorem does not yet produce three physical substances. The three-term structure enters only after a further, explicit construction: take a minimal normalized binary alternative with weights $1 - r$ and r , form its ordered product with itself, and remove order by exchange symmetry. The four ordered outcomes $(1 - r)^2$, $r(1 - r)$, $(1 - r)r$, r^2 compress into three exchange orbits:

$$(1 - r)^2, \quad 2r(1 - r), \quad r^2, \quad \text{with} \quad (1 - r)^2 + 2r(1 - r) + r^2 = 1.$$

The outer terms are the two pure pairings; the middle term is the total mixed relation. This is complete accounting, not yet cosmology. The labels *light*, *boundary*, and *matter* belong to a representation bridge; until that map is constructed, the partition is exact algebra and the physical reading is proposed.

2.3 Where the golden ratio enters

The fixed-point equation $\varphi^2 = \varphi + 1$ is not obtained merely by saying that a system describes itself. Under primitive whole-shift self-description (Anchor A1), the referent-code update is the initial

Fibonacci encoder, whose characteristic polynomial is $\lambda^2 - \lambda - 1$. The golden equation and the contraction coordinate

$$r = \frac{1}{2\varphi}, \quad 0 < r < \frac{1}{2},$$

then follow as theorems of that represented self-coding system. The remaining question is physical rather than algebraic: whether primitive self-description in Nature realizes Anchor A1, and whether the resulting invariant sectors are represented by the measured physical channels.

2.4 The whole chain in one view

Self-description motivates invariance. Termination plus local confluence proves unique normal forms. Normal forms define the semantic quotient. Binary execution and exchange symmetry produce the three-term partition. Under Anchor A1 the initial whole-shift encoder fixes φ and r . Under Anchors A2–A4 the partition receives a local dynamical representation, a conserved stress-energy readout, and a closed internal carrier. The cosmological formulas are therefore *conditional* consequences of a finite representation package, not unconditional consequences of rewriting alone.

The Golden Dictionary

For the golden branch define

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad r = \frac{1}{2\varphi}, \quad 0 < r < \frac{1}{2}.$$

The master construction assigns the following dimensionless quantities:

$p = 2r$	direct registration,
$h = 1 - 2r$	deferred registration,
$b = \frac{r^2}{2}$	baryonic fraction,
$d = 4r^2(1 - r)$	dark-matter fraction,
$e = 1 - b - d$	residual dark-energy fraction,
$q = r^3$	dimensionless action unit,
$\lambda = r^{240}$	dimensionless vacuum hierarchy.

On the golden fixed point,

$$p^2 + p = 1, \quad h = p^2.$$

3.1 Channel weights

The three exact execution weights of the exchange partition are

$$W_L = (1 - r)^2, \quad W_B = 2r(1 - r), \quad W_M = r^2, \quad W_L + W_B + W_M = 1,$$

read as light, boundary, and matter channels under the proposed physical identification.

3.2 Cosmological fractions

[Proposed / conditional under Anchors A1–A4]

Every cosmological density parameter is expressed as an explicit algebraic function of r :

$$\Omega_b := \frac{r^2}{2}, \quad \Omega_{\text{DM}} := 4r^2(1 - r), \quad \Omega_{\text{DE}} := 1 - \frac{9}{2}r^2 + 4r^3.$$

2026.2 CORRECTION (X1)

An earlier formula $\Omega_b = r^2(1 - r) \approx 0.0660$ conflicted with the form $\Omega_b = r^2/2 \approx 0.04775$ — one symbol, two inequivalent objects. This edition adopts $\Omega_b := r^2/2$ (residual 3.2% against the

observed 0.0493, versus 34% for the superseded form). Both remain graded *Proposed*: neither is derived, and the partition map connecting the channel weights to any Ω remains the smallest missing theorem of the cosmological bridge.

The phrase “zero-parameter cosmology” requires an honest qualification. Once these functions are fixed, nothing remains to tune, and each measured density becomes a yes/no test. It is *not* claimed that these are the only algebraic functions of r compatible with the axioms: infinitely many polynomials in r pass near any measured value, so agreement alone cannot establish the choice. What would establish it is a rigidity theorem.

CONJECTURE 3.1 Cosmological rigidity

Every cosmological invariant preserved by the execution algebra and compatible with the channel decomposition is a polynomial in r of degree bounded by its channel order, and the admissible polynomial is unique per invariant. *Status: Conjectured — no proof strategy is currently known.* Absent this theorem, the fractions above are the framework’s candidate dictionary, not its consequence.

3.3 The vacuum hierarchy

[Proposed]

The most dramatic entry in the dictionary is the vacuum energy density in Planck units,

$$\frac{\rho_\Lambda}{M_\star^4} \sim r^{240},$$

so that the observed hierarchy of roughly 10^{-122} appears as an exact power of the contraction rate. The candor required: the exponent 240 is at present an input matched to the observed hierarchy, not an output of the framework; the executed audit further showed the implied exponent shifts from 241.1 to 202.3 between Planck-mass conventions — a shift 39 times larger than the 240-versus-241 distinction itself. The section is retained because the re-expression is falsifiable in conjunction with the other sectors: *the same r must serve all sectors simultaneously*, which is a genuine constraint no single fitted formula would face. Part II makes that constraint exact.

The Representation Boundary

The bridge-closure programme reduced the physical content of the framework to four independent representation anchors, phrased as observable realization conditions rather than as unexplained numerical assignments, together with one dimensionful calibration.

ANCHOR A1 Primitive whole-shift self-description

Primitive physical self-description realizes the initial whole-shift system, and direct/deferred registration reads its persistent and generated stationary components.

ANCHOR A2 Conservative local monoidal realization

There is a conservative strong symmetric-monoidal physical representation of the behavioural execution category. It sends primitive independent completions to operationally independent systems, realizes the defect Hamiltonian locally, and has the generally covariant refinement limit. At the frozen internal point, the E_8 first-shell root factors form a monoidal family.

ANCHOR A3 Relational stress-energy and vacuum readout

The readout of physical records is the relational registration machine; execution throughput is converted to stress-energy with the common calibration; and residual vacuum stress-energy is M_*^4 times the unique maximal-defect projector.

ANCHOR A4 Closed spin outcome carrier

The compact internal code is a spin Poincaré pair whose middle homology is the minimal faithful carrier of the full three-coordinate outcome algebra A_3 . The effective theory is restricted to the first nonzero lattice shell.

The dimensionful calibration remains separate,

$$S_* r^3 = \hbar,$$

and is not an additional dimensionless structural choice.

THEOREM 4.1 Four-anchor master closure theorem

Assume the rewriting axioms R1–R4, Postulate A, Anchors A1–A4, and the single action calibration $S_* r^3 = \hbar$. Then the full downstream structure follows. Moreover:

1. the self-coding encoder is selected by an initial universal property;
2. rank eight is derived from the complete three-coordinate outcome algebra rather than assumed by minimality;
3. propagation, interface, and localization are operational spectral definitions;
4. the cosmological routing map is the unique pushforward of one relational registration machine;
5. equal channel-energy calibration is necessary and sufficient for the displayed density fractions;
6. the vacuum observable is the unique maximal-defect projector, and exact quadraticity is replaced by monoidality of the frozen root state.

The unconditional content is mathematical: normal forms, quotient factorization, universal constructions, explicit execution models, channel spectra, routing identities, coherence quantization, and the compact first-shell constructions proved under their stated hypotheses. The physical content is conditional and finite: Postulate A, Anchors A1–A4, and one dimensional action calibration specify which representation of that mathematics Nature is proposed to realize. The dependency structure is one chain rather than a collection of coincidences:

$$\begin{aligned}
 &\text{execution} \longrightarrow \text{behavioural quotient} \longrightarrow \text{whole-shift encoder} \\
 &\longrightarrow \text{channel tensor algebra} \longrightarrow \text{local continuum and readout} \\
 &\longrightarrow \text{compact carrier} \longrightarrow \text{frozen vacuum event.}
 \end{aligned}$$

Future progress can replace an anchor only by constructing and experimentally validating the corresponding physical representation; it cannot be achieved by silently relabelling an algebraic identity as a theorem.

Part II

Rigidity, Identifiability, and Falsification

STATUS OF PART II

Part II incorporates the 2026.8 Closing Addendum. It does not alter the coherent master results and it introduces no new physical bridge. Its purpose is narrower: to prove the exact relations that must hold *between* sectors once the master representation anchors are adopted, to distinguish genuine parameter freedom from representation freedom, and to state a compact falsification protocol. The results are algebraic consequences of the already-defined quantities and therefore provide an independent audit of whether the programme is behaving as one theory rather than as several unrelated fits.

The Remaining Itch: Unity Must Mean Overdetermination

A framework is not unified merely because the same symbol appears in several formulas. The same invariant must be recoverable independently from several sectors, and those recoveries must agree. Otherwise the symbol can silently become a different fitted parameter in every application.

Recall the golden dictionary of Chapter 3: the registration pair (p, h) , the routed background fractions (b, d, e) , the action unit $q = r^3$, and the vacuum hierarchy $\lambda = r^{240}$, all built from the single contraction $r = 1/(2\varphi)$.

The question of this part is simple: *how many independent numbers are these?* The answer is one. More strongly, any one of several sectors determines all the rest.

Exact Cross-Sector Elimination

THEOREM 6.1 Cross-sector closure identities

Assume the golden branch and the routed quantities of Chapter 3. Then the following exact identities hold:

$$\begin{aligned} h &= 8b, & q^2 &= 8b^3, & \lambda &= q^{80} = (2b)^{120}, \\ d &= 8b(1 - \sqrt{2b}), & e &= 1 - b - d, \end{aligned}$$

and, eliminating the square root,

$$(8b - d)^2 = 128b^3, \quad 0 < d < 8b.$$

Proof. Since $p = 2r$ and $h = p^2$, one has $h = 4r^2 = 8b$. Since $q = r^3$ and $b = r^2/2$,

$$q^2 = r^6 = (r^2)^3 = (2b)^3 = 8b^3.$$

Likewise

$$\lambda = r^{240} = (r^3)^{80} = q^{80} = (r^2)^{120} = (2b)^{120}.$$

For dark matter, $d = 4r^2(1 - r) = 8b(1 - \sqrt{2b})$, because $r = \sqrt{2b}$ on the physical branch. Hence $8b - d = 8b\sqrt{2b}$, and squaring gives $(8b - d)^2 = 128b^3$. The sign condition $0 < d < 8b$ selects the physical square-root branch. The residual identity is its definition. ♦

COROLLARY 6.2 Registration alone fixes the background dictionary

If p is measured and the golden normalization $p + p^2 = 1$ holds, then

$$r = \frac{p}{2}, \quad b = \frac{p^2}{8}, \quad h = p^2, \quad q = \frac{p^3}{8}, \quad \lambda = \left(\frac{p}{2}\right)^{240},$$

and

$$d = p^2 \left(1 - \frac{p}{2}\right), \quad e = 1 - b - d.$$

COROLLARY 6.3 Baryonic fraction fixes the hierarchy

On the physical branch, $r = \sqrt{2b}$ and therefore

$$\frac{\rho_\Lambda}{M_*^4} = (2\Omega_b)^{120}$$

under the master vacuum identification. The baryonic routing fraction and the vacuum hierarchy are not independent assignments inside the completed representation.

6.1 Channel weights in the same coordinate

The three exact execution weights can be written entirely in terms of b :

$$W_M = 2b, \quad W_B = 2\sqrt{2b}(1 - \sqrt{2b}), \quad W_L = (1 - \sqrt{2b})^2,$$

and they satisfy $W_L + W_B + W_M = 1$. Hence a single routed background quantity fixes the underlying channel simplex as well.

Independent Estimators of the Same Invariant

A sharp theory should permit the same invariant to be reconstructed in several ways. Define the sector estimators

$$\hat{r}_{\text{reg}} = \frac{\hat{p}}{2}, \quad \hat{r}_b = \sqrt{2\hat{b}}, \quad \hat{r}_q = \hat{q}^{1/3}, \quad \hat{r}_\Lambda = \hat{\lambda}^{1/240}.$$

For dark matter define $\hat{r}_d$ as the unique solution in $(0, \frac{1}{2})$ of

$$4r^2(1 - r) = \hat{d}.$$

THEOREM 7.1 Uniqueness of the dark-matter estimator

The map $f(r) = 4r^2(1 - r)$ is strictly increasing on $(0, \frac{1}{2})$. Therefore every d in the image of this interval determines a unique r_d .

Proof. $f'(r) = 8r - 12r^2 = 4r(2 - 3r) > 0$ for $0 < r < \frac{1}{2}$. ◆

THEOREM 7.2 Diagonal consistency criterion

Let

$$\hat{\mathbf{r}} = (\hat{r}_{\text{reg}}, \hat{r}_b, \hat{r}_d, \hat{r}_q, \hat{r}_\Lambda).$$

The cross-sector realization of Unified Mechanics is consistent exactly when the admissible uncertainty regions for these estimators possess a common intersection. In the zero-uncertainty idealization this is

$$\hat{r}_{\text{reg}} = \hat{r}_b = \hat{r}_d = \hat{r}_q = \hat{r}_\Lambda.$$

This is the operational content of using one r . Allowing a separate value in each sector replaces one theory by five unrelated one-parameter descriptions.

PROPOSITION 7.3 Codimension of unification

If n sectors are each allowed an independent effective contraction r_i , their unconstrained parameter space is $(0, \frac{1}{2})^n$. The unified condition

$$r_1 = r_2 = \cdots = r_n$$

defines the diagonal, a one-dimensional submanifold of codimension $n - 1$.

Thus every additional independently readable sector adds a genuine consistency constraint rather than an additional fitting freedom.

7.1 A normalized residual vector

For practical comparison define a reference estimator $\bar{r}$ and normalized residuals

$$Z_i = \frac{\hat{r}_i - \bar{r}}{\sigma_i},$$

where σ_i is the uncertainty propagated into the r coordinate. A natural weighted common estimator is

$$\bar{r} = \frac{\sum_i \hat{r}_i / \sigma_i^2}{\sum_i 1 / \sigma_i^2}.$$

The statistic

$$\chi_{\text{UM}}^2 = \sum_i Z_i^2$$

is not itself a theorem of the framework, but it is the direct empirical test of its single-invariant claim once independent measurements and covariance information are supplied.

Parameter-Free Does Not Mean Representation-Free

The master edition separates proved mathematics from physical representation anchors. One final distinction is useful: an anchor need not be a numerical fit parameter, but it can still contain a family of inequivalent physical representations.

Let $\mathfrak{R}$ be the class of representations satisfying the stated anchors, modulo the equivalence relation regarded as physically irrelevant. Let

$$\mathcal{O} : \mathfrak{R} \rightarrow Y$$

be a predicted observable.

THEOREM 8.1 Representation-moduli criterion for derivation

The observable $\mathcal{O}$ is rigidly derived from the stated axioms and anchors only if it is constant on $\mathfrak{R}$. If two admissible representations $R_1, R_2 \in \mathfrak{R}$ satisfy all premises but

$$\mathcal{O}(R_1) \neq \mathcal{O}(R_2),$$

then $\mathcal{O}$ is not yet uniquely derived.

Proof. A logical consequence must hold in every model of its premises. Two models of the premises with different values of $\mathcal{O}$ are therefore a countermodel to uniqueness. $\blacklozenge$

COROLLARY 8.2 Honest zero-parameter criterion

A sector is zero-free-parameter in the strong physical sense only when its admissible representation moduli leave all claimed dimensionless observables invariant. Discrete structural anchors do not count as fitted numbers, but unclassified functional freedom inside an anchor cannot be ignored.

8.1 Background rigidity versus perturbation freedom

The routed background fractions are rigid after the relational machine and common energy calibration are fixed. The perturbation sector is different.

THEOREM 8.3 A fixed background does not determine perturbations

Let $P_0(X, \phi)$ be a scalar-field Lagrangian admitting a background trajectory $X = X_0(\phi)$. For any sufficiently regular function $\alpha(\phi)$ define

$$P_\alpha(X, \phi) = P_0(X, \phi) + \alpha(\phi)(X - X_0(\phi))^2.$$

Along the background trajectory, P_α , its first X derivative, and its background variation agree with those of P_0 , so the same background solution is retained. But

$$(P_\alpha)_{XX}|_{X_0} = (P_0)_{XX}|_{X_0} + 2\alpha(\phi),$$

so the scalar sound speed

$$c_s^2 = \frac{P_X}{P_X + 2XP_{XX}}$$

generally changes with α . Therefore a background equation of state does not uniquely determine perturbation propagation.

Proof. The added term and its first X derivative vanish at $X = X_0(\phi)$. Its derivative along the specified trajectory also vanishes because $X - X_0(\phi)$ vanishes identically there. The second X derivative does not vanish and equals 2α . The displayed sound-speed formula therefore changes unless the deformation is dynamically irrelevant. ◆

This theorem fixes the proper scope of the parameter claim:

- the golden algebra, channel weights, routed background fractions, dimensionless r^3 lattice, and conditional r^{240} weight are rigid under the master constructions;
- a unique continuum perturbation theory still requires the physical representation in Anchor A2 to select a specific operator, not merely a background history.

A Compact Falsification Ledger

The framework can be tested without repeatedly solving for r . The following identities eliminate it entirely.

Bridge compared	Exact relation	Failure means
Direct / deferred registration	$p + h = 1, h = p^2$	The stationary golden registration model fails.
Registration / baryons	$h = 8b$	Registration and routed matter do not share one r .
Baryons / dark matter	$(8b - d)^2 = 128b^3, 0 < d < 8b$	The proposed conserved routing map fails.
Background closure	$b + d + e = 1$	The three-output accounting is incomplete or misidentified.
Baryons / action lattice	$q^2 = 8b^3$	The r^3 coherence unit is not the same contraction as matter routing.
Action / vacuum	$\lambda = q^{80}$	The 240-root vacuum event is not built from the same three-channel unit.
Baryons / vacuum	$\lambda = (2b)^{120}$	The root-shell hierarchy and cosmological routing are inconsistent.
All readable sectors	$r_{\text{reg}} = r_b = r_d = r_q = r_{\Lambda}$	The framework has split into sector-specific effective parameters.

Table 9.1: The falsification ledger: every row is an r -free identity whose failure has a named meaning.

THEOREM 9.1 No rescue by silent refitting

If one of the exact elimination identities fails and the failure is repaired only by assigning a distinct contraction parameter to that sector, the repaired model is not the same unified theory. It is an extension with at least one additional independent parameter and must be graded as such.

This is not a rhetorical rule. It follows from Proposition 7.3: leaving the diagonal enlarges the model class.

9.1 Hierarchy sensitivity

PROPOSITION 9.2 Vacuum hierarchy sensitivity

For $\lambda = r^{240}$,

$$\frac{d \log \lambda}{d \log r} = 240.$$

Hence a small fractional displacement obeys

$$\frac{\delta \lambda}{\lambda} \approx 240 \frac{\delta r}{r}.$$

The hierarchy is therefore an exceptionally sharp cross-sector test — but only after the same dimensionless normalization and physical vacuum observable have been fixed.

The Exact Golden Benchmark

The exact branch gives

$$\varphi = 1.618033988749894848 \dots, \quad r = 0.309016994374947424 \dots$$

Quantity	Golden value
Direct registration p	0.618033988749894848 ...
Deferred registration h	0.381966011250105152 ...
Baryonic fraction b	0.047745751406263144 ...
Dark-matter fraction d	0.263932022500210304 ...
Residual fraction e	0.688322226093526552 ...
Dimensionless action unit $q = r^3$	0.029508497187473712 ...
Vacuum hierarchy $\lambda = r^{240}$	$3.9424607712196638 \times 10^{-123}$
$\log_{10} \lambda$	-122.404232619350383 ...
Light-channel weight W_L	0.477457514062631440 ...
Boundary-channel weight W_B	0.427050983124842272 ...
Matter-channel weight W_M	0.095491502812526288 ...

Table 10.1: The exact golden benchmark. These numbers are not independent successes; they are one point on the cross-sector consistency manifold. The scientific claim is the manifold, not the length of the table.

Closing Statement

The final structural lesson is this:

one invariant is meaningful only when many sectors
cannot choose it independently

The coherent master edition closes the dependency architecture under explicit representation anchors. Part II closes the corresponding *identifiability* architecture. It shows exactly how the same contraction must reappear in registration, channel accounting, cosmological routing, coherence quantization, and the root-shell hierarchy. It also proves where the claim must stop: a background history does not uniquely fix perturbations, and a representation anchor is not equivalent to a theorem merely because it contains no fitted decimal.

The programme therefore ends this session with both sides exposed:

- a strongly overdetermined dimensionless core, with exact elimination identities and multiple independent estimators of r ;
- a finite physical representation boundary, whose remaining continuum and semantic content must be constructed or tested rather than rhetorically absorbed into the algebra.

That is the cleanest form of the claim. It is stricter than saying that many formulas contain the golden ratio, and stronger than treating every agreement separately. *Unified Mechanics succeeds or fails as one object.*



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